SmartSupport AI

An Agentic AI-Powered E-Commerce Customer Support Platform

Prepared by :

Salma Mohamed Hafez Mahmoud

Engy Abdelrahman Mohamed Nagy

Ali Mohamed Abdelmawgoud Nassef

Khaled Osama Abdellatef Selem

Zeyad Mohamed Samir Mohamed

1. Project Information

Project Type: Full Stack .NET Web Development

Domain: E-Commerce, Customer Support Automation, and Artificial Intelligence

Team Size: 5 Members

Estimated Duration: 6 To 8 Weeks

Primary Focus: Backend Engineering, Agentic AI, Workflow Automation, and Database Management

2. Executive Summary

As e-commerce businesses grow, managing customer inquiries about orders, deliveries, returns, refunds, and account issues becomes increasingly challenging. Traditional support channels and rule-based chatbots often struggle to understand complex requests, access customer-specific information, execute business operations, and maintain context across conversations. Consequently, support teams face repetitive workloads, delayed responses, and inefficient escalation processes.

SmartSupport AI is an intelligent, Agentic AI-powered customer support platform designed to transform how e-commerce businesses manage customer interactions. Unlike conventional chatbots that primarily provide predefined answers, SmartSupport AI combines Large Language Models (LLMs), knowledge retrieval, intent recognition, and controlled tool execution to help customers resolve issues through natural conversations.

The platform will enable customers to ask questions, check order statuses, submit refund requests, update permitted account information, and create support tickets through an intelligent chat interface. The AI agent will retrieve relevant information from an approved knowledge base, invoke authorized backend operations, and maintain conversational context throughout the interaction.

When an inquiry requires human judgment, involves a sensitive operation, or exceeds the AI's capabilities, the system will escalate it to a human support agent with a summarized conversation, relevant customer information, and an explanation of the escalation reason.

3. Problem Statement

E-commerce businesses commonly encounter the following customer support challenges:

- **Repetitive inquiries:** Customers frequently ask about order status, shipping policies, returns, refunds, and product information.
- **Slow response times:** Support teams may be overwhelmed by incoming requests, particularly during sales and promotional periods.
- **Limited chatbot capabilities:** Traditional chatbots often rely on predefined rules and cannot effectively manage contextual conversations or multi-step operations.
- **Disconnected systems:** Customer information, order records, support conversations, and tickets may be stored in separate systems.
- **Inefficient escalation:** Human agents may receive incomplete context when a conversation is transferred from an automated assistant.
- **Inconsistent support quality:** Responses may vary across agents, channels, and customer interactions.
- **Limited performance visibility:** Businesses need to understand resolution rates, response times, escalation patterns, customer satisfaction, and support workload.

These challenges demonstrate the need for an integrated solution that combines conversational intelligence with reliable backend operations, structured workflows, and human supervision.

4. Proposed Solution

SmartSupport AI will provide a centralized support platform connecting customers, AI agents, human support representatives, and administrators through a single application.

The system will follow an **AI → validated backend tool → business operation → recorded result** workflow.

Rather than granting the language model unrestricted database access, the application will expose a controlled set of backend tools. The AI will select an appropriate tool based on the customer's request, while the backend remains responsible for authorization, validation, business rules, transaction management, and data integrity.

The platform will provide three complementary capabilities:

4.1 Intelligent Conversational Support

The AI assistant will understand customer intent, answer questions using approved knowledge sources, maintain conversation context, and provide responses in the customer's detected language where supported.

4.2 Controlled Business-Process Automation

The AI agent will invoke predefined backend tools to retrieve order information, initiate refund workflows, submit account updates, schedule callbacks, and create support tickets.

Sensitive actions will require the appropriate authorization and, where necessary, explicit customer confirmation or human approval.

4.3 Human-AI Collaboration

Complex or sensitive cases will be transferred to a human agent. The agent will receive the conversation history, an AI-generated summary, relevant customer context, and the reason for escalation.

Human agents will also be able to use AI-generated response suggestions without surrendering control over the final response.

5. Project Objectives

The project aims to:

1. Build a maintainable e-commerce customer support platform using C# and ASP.NET Core.
2. Demonstrate ASP.NET Core MVC, RESTful Web APIs, Entity Framework Core, and SQL Server.
3. Implement a modular, layered architecture using established software design patterns.
4. Integrate a free-tier LLM API, where available, through an abstraction that supports replacing the provider.
5. Develop an AI agent capable of intent recognition, contextual conversations, knowledge retrieval, and controlled tool calling.
6. Implement a knowledge base for FAQs, shipping policies, return policies, and other support documentation.
7. Connect AI tools to validated backend operations involving orders, customer information, and support requests.
8. Develop real-time customer messaging using SignalR.

9. Implement intelligent escalation, ticket management, and human-agent collaboration.
10. Provide administrative controls and analytics for evaluating support operations and AI performance.
11. Apply authentication, role-based authorization, input validation, transaction management, audit logging, and secure configuration.
12. Use Git/GitHub and Docker to support collaboration, reproducible development, and deployment preparation.

6. Target Users and Business Value

Customers

- Receive support at any time through a conversational interface.
- Check order information without waiting for an agent.
- Obtain answers grounded in approved company policies.
- Submit support requests and track their progress.
- Escalate issues to human representatives when necessary.

Customer Support Agents

- Manage assigned conversations and escalated cases.
- Access customer and order context.
- Review AI-generated conversation summaries.
- Use suggested responses to reduce repetitive work.
- Manage tickets, priorities, and resolution statuses.

Support Managers

- Monitor support performance, response times, resolution rates, and escalation patterns.
- Identify recurring customer problems.
- Evaluate customer satisfaction and AI-assisted workflows.

System Administrators

- Manage users, roles, agents, and knowledge articles.
- Configure AI prompts and system settings.
- Review audit records and monitor application performance.

Primary business value: SmartSupport AI aims to reduce repetitive support work, shorten customer waiting times, improve consistency, and give businesses better visibility into their support operations.

7. Technical Approach

SmartSupport AI will use a modular, layered architecture combining .NET backend development with Agentic AI capabilities.

Programming Languages and Frameworks: C#, ASP.NET Core MVC, and ASP.NET Core Web API will build the application and RESTful endpoints. HTML, CSS, JavaScript, and Bootstrap 5 will support the interface, while SignalR will enable real-time communication.

Backend Architecture: N-Tier Architecture will separate presentation, business logic, AI services, and data access. Repository, Unit of Work, Chain of Responsibility, Strategy, and Dependency Injection patterns will improve maintainability and testability.

Database and Security: SQL Server will store customers, orders, conversations, tickets, and knowledge articles. Entity Framework Core will manage data access, while ASP.NET Identity will provide authentication and role-based authorization.

Artificial Intelligence: A free-tier LLM API, where available, will power conversational responses and tool calling. Microsoft Semantic Kernel will support AI orchestration. Prompt engineering, intent recognition, sentiment analysis, and Retrieval-Augmented Generation (RAG) will enable contextual, knowledge-grounded support.

APIs and Knowledge Retrieval: The system will retrieve relevant support articles and expose controlled tools for order-status checks, refund requests, customer updates, and ticket creation, with server-side validation and authorization.

Development and Deployment: Git/GitHub will support collaboration, Docker will provide containerization, and automated tests will validate critical workflows. Microsoft Azure services may support deployment, monitoring, and knowledge search, depending on availability and cost.